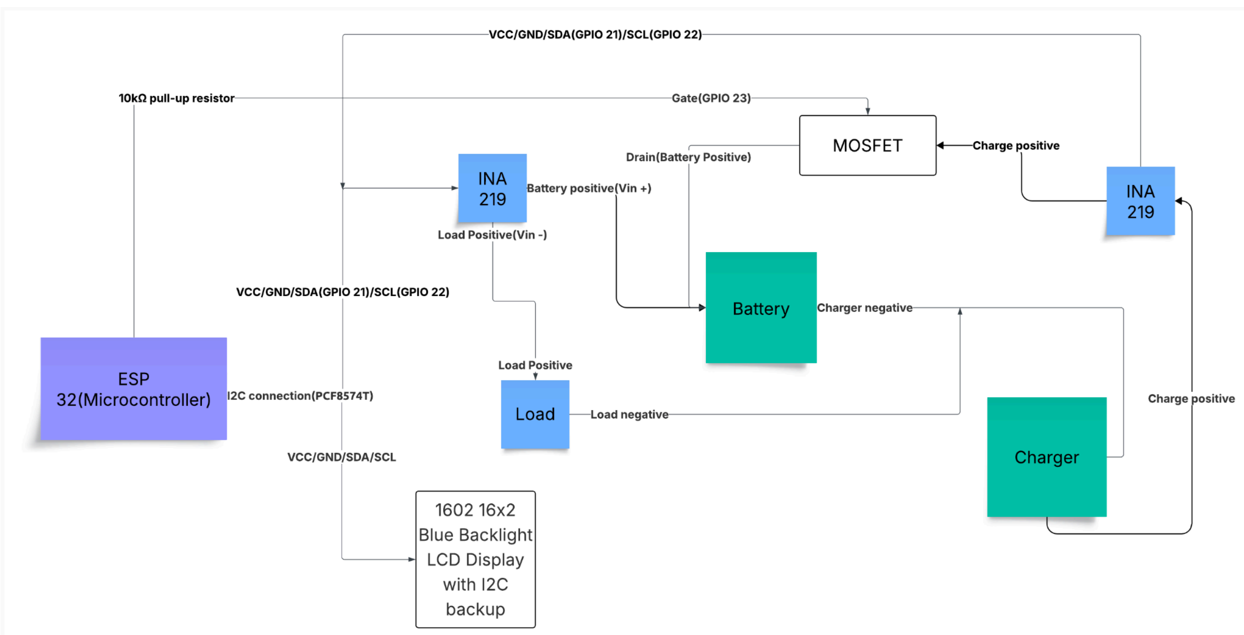
Updated Project Proposal.

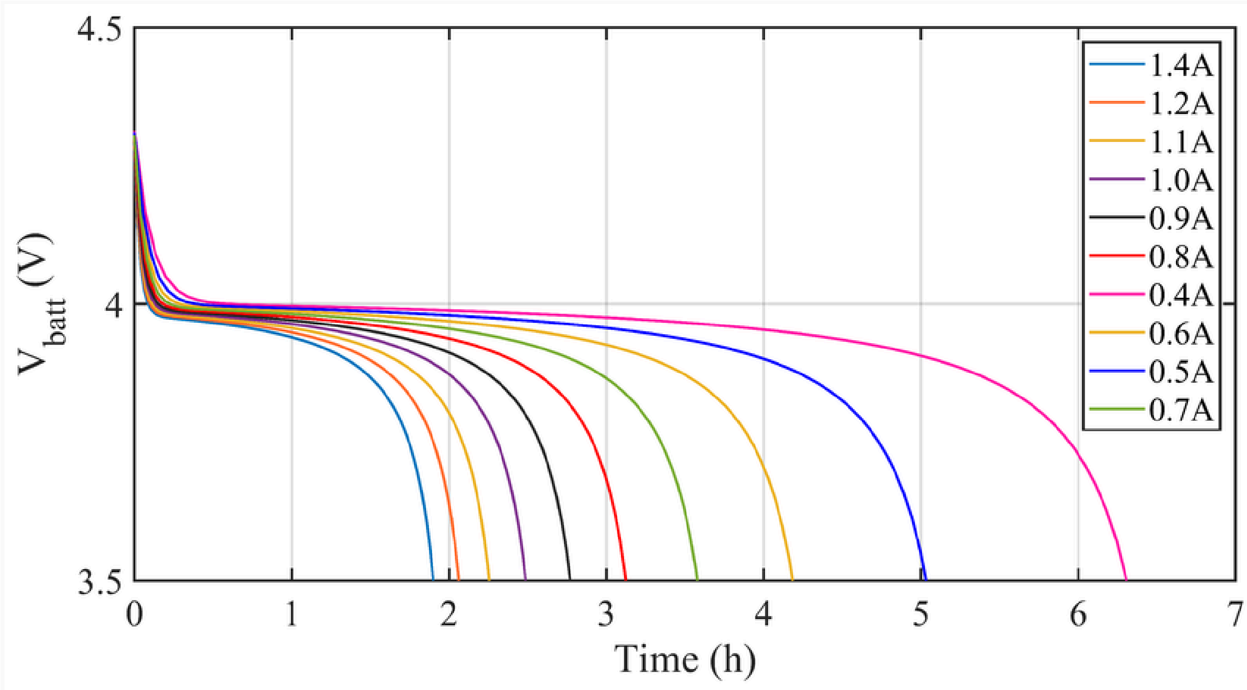
Relevant Components :
INA219 Current Monitor Voltage Sensor Module I2C 0-26V 3.2A Supply 3-5VDC - Rs.420
https://tronic.lk/product/ina219-current-monitor-voltage-sensor-module-i2c-0-26v-?srsltid=AfmBOop3T3LZ8Y2lJewfmsqibkeW2jtlms9QVPsp0FdidRe1dPagNcir
IRF540N Power MOSFET - Rs.70
https://tronic.lk/product/irf540n-power-mosfet?srsltid=AfmBOoqewqEqIg4nlYBA_cxQlp8QPsZecLCFkeRiac2KBTwUYFUJ3uXc
Battery Charger TP4056- Rs.50
https://tronic.lk/product/tp4056-5v-1a-micro-usb-18650-special-lithium-battery-ch?srsltid=AfmBOop14VVq15i8IMJCYJFbC5WOzY3hCpCEa9Cz5Cdsie217vtOhrsM
Blue pill board - Rs.550
https://tronic.lk/product/stm32f103c8t6-stm32-arm-minimum-system-development-boar?srsltid=AfmBOoprMcweLv4d67cCwJYlzP8xv5Htqe_aBLvBzJbg3_WRiSwRcjtS
Buck Converter
https://tronic.lk/product/lm2596s-3-40v-to-1-5-35v-4a-dc-to-dc-adjustable-step-do?srsltid=AfmBOor5tCZxJEMmgpXjWGWy105Qn0zu9uaE82mLJSbTTBetTbD8Fonr

Hardware Structure :

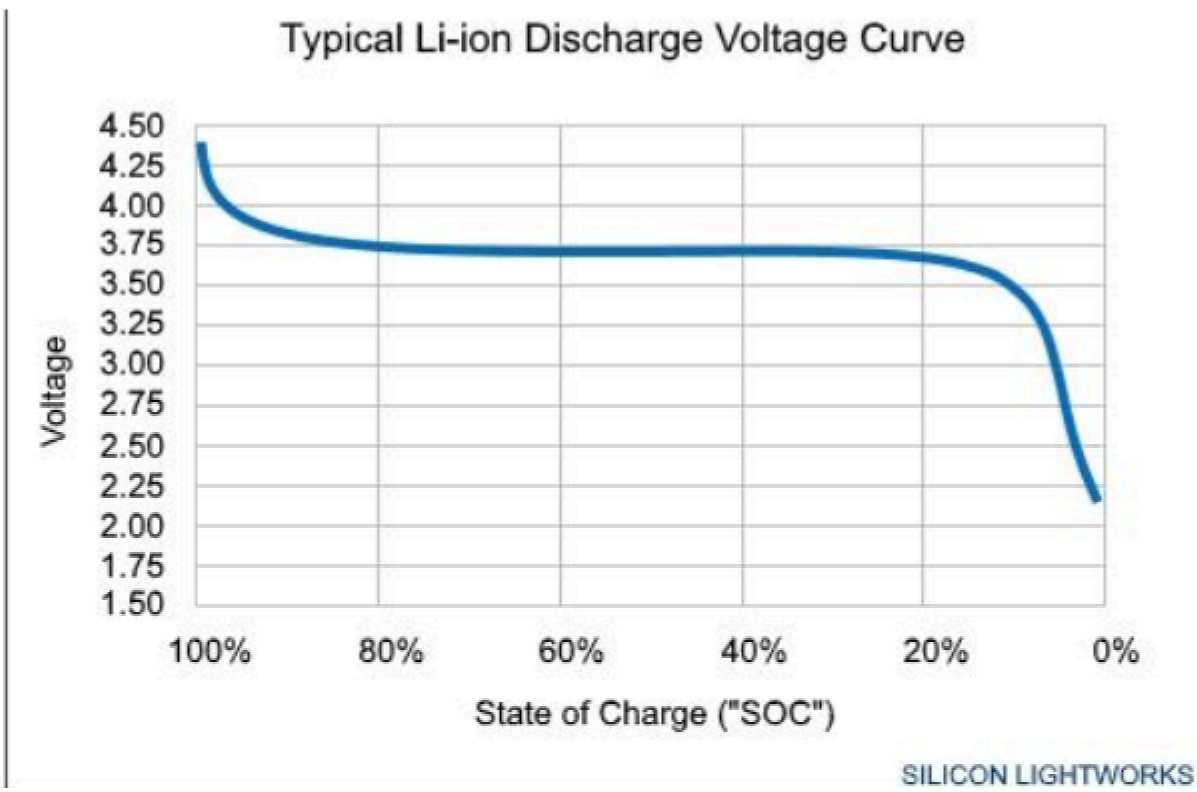


Link :
https://lucid.app/lucidchart/ab63548f-051f-4156-9031-f49bf163c6a3/edit?viewport_loc=-1075%2C-928%2C2844%2C1316%2C0_0&invitationId=inv_29a397d6-b607-4a47-abae-1adfe2832558

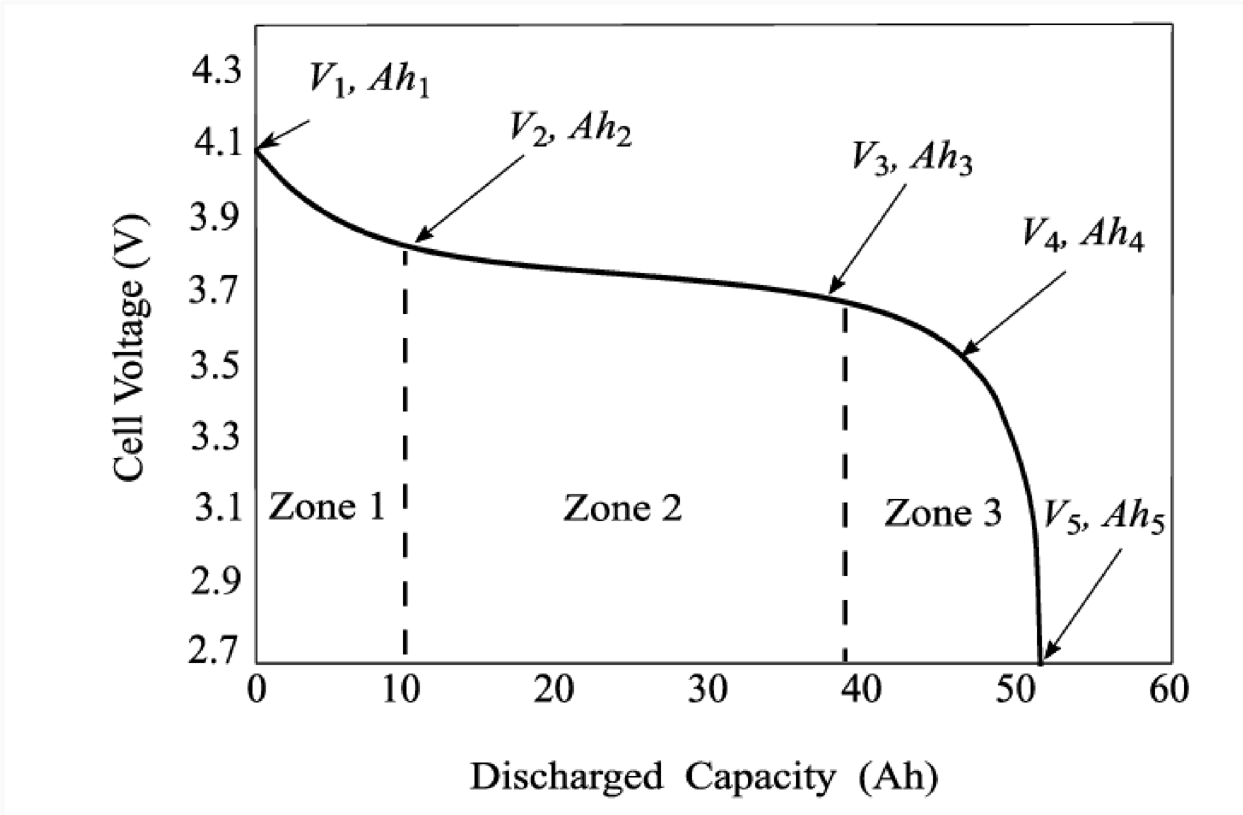
Graph analysis and algorithms :



Graph 1

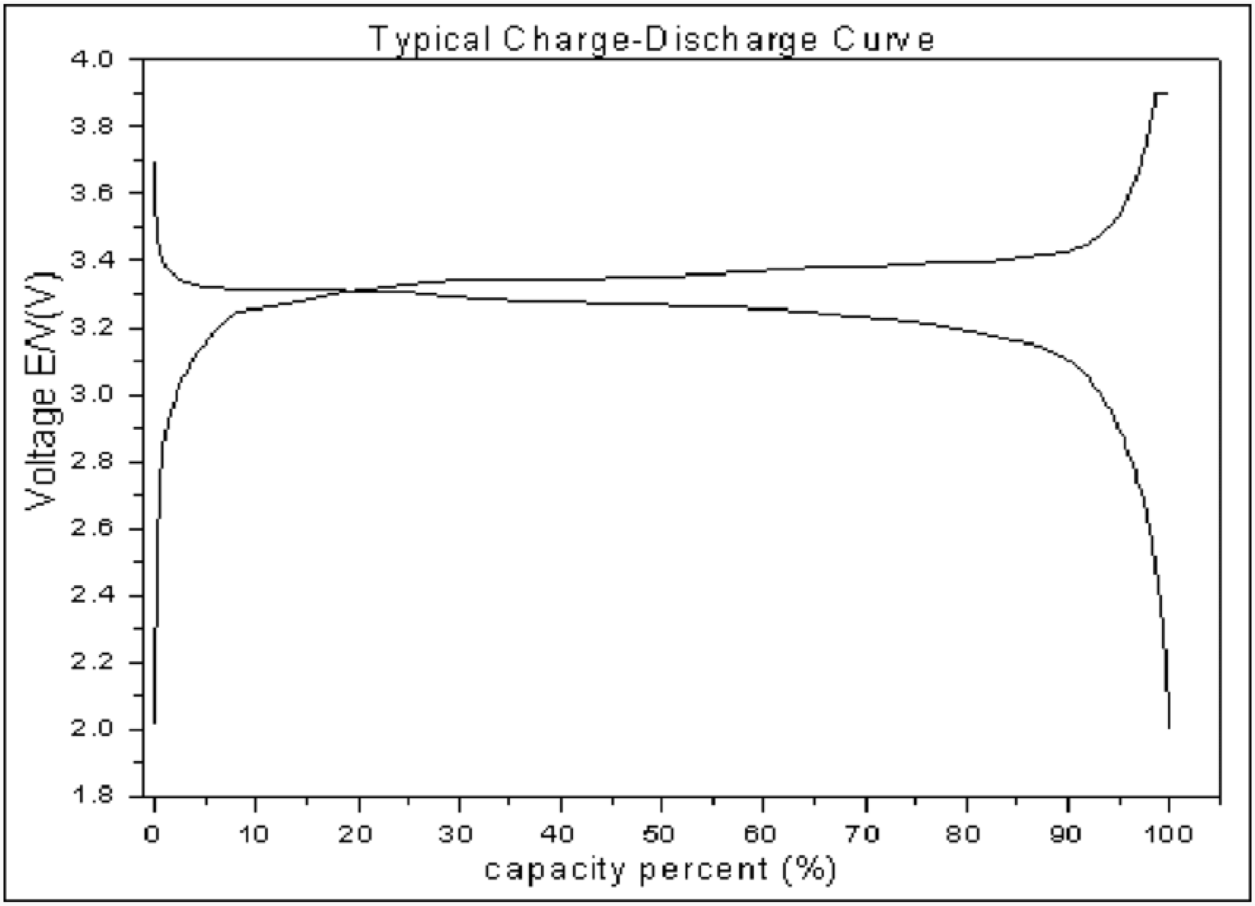


Graph 2



4.3
4.1
3.9
3.7
3.5
3.3
3.1
2.9
2.7
Cell Voltage (V)
0
10
20
30
40
50
60
Discharged Capacity (Ah)
V1, Ah1
V2, Ah2
V3, Ah3
V4, Ah4
V5, Ah5
Zone 1
Zone 2
Zone 3

Graph 3



Graph 4

Process :

- Power up the battery and calculate the voltages and current draw via INA module and use a timer in esp 32 to get the time.
- Then it breaks the connection with the charger and begins to power up the load.
- The INA module connected to the load will give the current draw and voltage of the system.
- Using a calculation the remaining time and SoC will be displayed in the LED screen.
- And give alert when the battery’s voltage is 3 v when discharging.(at 3V the battery capacity will be 5% of total capacity.For li-ion battery which is powered by 3200 mAh it would be 160mAh).
- When charging state give an alert at 3.8V

Used algorithms

The charging curve and the discharge curve with capacity vary in nearly the same manner(Graph 4).

- Capacity(mAh) = Current(mA) x time
- Time = Capacity / Current

Future improvements

- Using simple machine learning model to get the output without charging inputs

Comments and Suggestions

- Week 1
  Change the battery management system to energy management system
  Try to charge the cells when discharging at the special position
- Week 2
  Understand the diagrams
  Use an AI model and algorithms
- Week 3
  No need to use AI model
  Reduce the components(High cost and high power consumption)
  Restructure the structures
- Week 4
  Refer the graphs and simplify the system
- Week 5
  Use data sheets to help
  Try to find good MOSFET
  And more efficient battery charger than TP4036
  Add the MOSFET near the charger.
- Week 6
  Draw a circuit diagram
  Change the mosfet to an manual switch
- Week 7
  Using a blue pill board(STM32F103C8T6 STM32 ARM Minimum System Development Board Blue Pill)to reduce the power consumption and working in low voltages.
  Plan to use a buck converter to make blue pill working in a 4V system.